

Luis Fernando Méndez López

Data Engineer

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Summary

Data Engineer experienced in building Python and SQL data pipelines that integrate high-volume external sources, validate them automatically and turn them into reliable, reproducible deliverables. Skilled in lakehouse architecture, dimensional modeling, data quality, PySpark, Delta Lake and dbt.

Experience

Sandoz, Business Intelligence Analyst Intern

May 2025 - Present

- Built a Python pipeline that processes the monthly **IQVIA DDD, PMM, NRC and PBS** audits (up to 13M records), cutting processing time from **6 hours to 10 minutes** and getting territory performance data to the sales force sooner, with **zero differences**.
- Automated ingestion of the raw IQVIA file (approx. 930 MB) with control-total validation, preventing source errors from double-counting sales in commercial reports.
- Integrated the **Close-Up and PBS** prescription audits (1.6M records, 24 months), matching physicians with 85% coverage so the commercial team could compare both sources at physician level.
- Modeled the impact of **drug samples** on prescriptions using a 2.36M-observation panel (fixed effects, PSM, LightGBM + SHAP) and designed a same-budget reallocation projecting **+55%** incremental prescriptions at no extra cost.
- Performed a **market sizing** analysis integrating sales (PMM, NRC) and prescription (PBS) data, revealing a 2.6x larger addressable market than initially scoped; delivered an interactive Excel model.
- Built Power BI dashboards used by sales managers and reps to track quotas, sales and opportunities by territory.
- Supported CRM (Salesforce) data governance, ensuring data quality, consistency and traceability.

Independent Contractor, Data Engineer / Database Administrator

Jul 2025 - Dec 2025

- Designed a centralized relational database to consolidate construction estimates from non-standardized sources, with automated ETL for cleansing and validation.
- Modeled tables with keys and referential integrity as the single source of truth for cost control and project progress.
- Optimized queries and reporting views, reducing extraction time.

Project

E-commerce Data Platform (Olist)

Personal project, 2026

Python, PySpark, Delta Lake, dbt, PostgreSQL, Docker, Airflow, MinIO (S3)

- Medallion lakehouse (Bronze/Silver/Gold) on the real-world Olist dataset (9 tables), with idempotent ingestion into partitioned Parquet and decisions documented in ADRs.
- Silver layer in PySpark and Delta Lake: 8 entities with deduplication and primary-key validation.
- Kimball star schema (5 dimensions, 4 facts) with a Type 2 SCD loaded via Delta MERGE, correctly attributing 100% of historical GMV.
- dbt semantic layer with 4 marts and 30 data tests; platform containerized with Docker Compose.

Skills

Languages: Python, SQL

Data: PySpark, pandas, Delta Lake, Parquet, PostgreSQL, dbt, Airflow

Concepts: ETL/ELT, dimensional modeling, SCD Type 2, data quality

Cloud & Tools: Azure (fundamentals), S3 / MinIO object storage, Docker, Git, Power BI, Salesforce

Data Quality: Control-total reconciliation, referential integrity, entity resolution (fuzzy matching)

Education

B.S. Industrial Engineering, National Autonomous University of Mexico (UNAM)

2021 - 2026

Certifications & Languages

Data Analyst Associate and Data Analyst in Power BI (DataCamp); SQL (University of Colorado Boulder); Applied Machine Learning in Python (University of Michigan); Initiating and Planning Projects (UC Irvine); Excel Skills for Business (Macquarie University).

Spanish (native) | English (upper-intermediate)